

TROY, AL, USA

WMO: 722267

Lat: 31.861N

Lon: 86.012W

Elev: 121

StdP: 99.88

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 03878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.6	-12.4	1.3	0.9	-9.8	1.7	3.3	9.2	8.1	8.4	8.7	1.4	330	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.1	24.0	34.0	24.1	33.0	24.1	26.2	31.5	25.7	31.0	25.3	30.5	2.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.2	27.9	24.3	19.5	27.5	23.9	19.0	27.2	81.5	31.4	79.4	31.1	77.5	30.6	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.9	5.7	DB	-8.1	36.4	2.2	1.7	-9.7	37.7	-11.0	38.7	-12.2	39.7	-13.8	41.0	
			WB	-8.8	27.1	2.2	0.7	-10.4	27.7	-11.7	28.1	-13.0	28.5	-14.6	29.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	8.6	10.8	14.3	18.2	22.3	25.9	27.0	26.9	24.6	18.9	13.2	10.1
	DBStd	7.69	5.64	5.07	4.80	3.67	3.02	1.75	1.42	1.71	2.73	4.29	4.67	5.24
	HDD10.0	250	97	47	17	1	0	0	0	0	1	19	67	
	HDD18.3	1185	304	214	140	47	7	0	0	2	46	164	259	
	CDD10.0	3332	52	71	151	246	382	478	528	524	438	277	116	70
	CDD18.3	1226	2	5	15	42	131	228	269	266	189	64	11	4
	CDH23.3	11319	5	23	130	434	1278	2182	2558	2429	1622	569	76	13
(13)	CDH26.7	4844	0	1	14	86	487	1040	1218	1142	693	154	8	0
(14)	Wind	WSAvg	2.1	2.6	2.7	2.8	2.4	2.0	1.7	1.6	1.8	2.0	2.1	2.4
(15) Precipitation	PrecAvg	1415	125	123	160	111	92	128	152	114	95	73	106	117
	PrecMax	2308	272	302	390	233	265	321	396	286	338	310	268	414
	PrecMin	1031	24	22	10	18	3	26	42	14	16	0	30	31
	PrecStd	226	59	67	83	61	54	72	78	61	64	69	48	66
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	25.9	28.3	30.2	33.6	36.1	36.1	37.1	35.1	31.7	27.7	25.2
		MCWB	18.3	19.1	18.3	20.4	22.2	23.5	24.3	24.3	23.4	22.8	20.2	18.9
	2%	DB	21.9	23.6	26.6	28.8	32.2	34.6	34.7	34.9	33.5	29.9	25.6	22.8
		MCWB	17.4	17.4	18.2	19.8	21.8	23.5	24.5	24.4	23.4	21.2	19.0	18.6
	5%	DB	20.1	21.5	24.7	27.6	31.0	33.4	33.5	33.5	32.3	28.4	23.7	21.1
		MCWB	16.7	16.7	17.2	19.1	21.6	23.7	24.6	24.7	23.3	20.5	18.0	17.9
	10%	DB	17.9	19.8	22.8	26.2	29.6	32.2	32.4	32.3	30.9	26.9	22.2	19.1
		MCWB	15.0	15.3	16.1	18.2	20.9	23.3	24.5	24.5	22.9	19.7	17.4	16.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.6	20.7	23.0	24.3	26.3	26.8	26.9	25.7	24.7	22.8	21.4
		MCDB	21.7	23.7	24.2	27.5	30.0	31.6	32.1	31.9	30.9	29.0	25.3	23.3
	2%	WB	18.8	19.3	19.8	21.8	23.5	25.6	26.1	26.3	25.1	23.6	21.4	19.9
		MCDB	21.0	22.1	23.3	25.9	28.8	31.2	31.6	31.4	30.2	27.0	23.3	21.6
	5%	WB	17.4	18.1	19.0	20.8	22.9	25.0	25.6	25.7	24.6	22.7	19.9	18.5
		MCDB	19.1	20.5	22.8	24.8	28.0	30.6	31.1	30.8	29.5	25.8	22.5	20.1
	10%	WB	15.7	16.7	17.8	19.8	22.3	24.4	25.2	25.3	24.0	21.6	18.4	16.9
		MCDB	18.1	19.0	21.4	23.7	27.2	29.5	30.5	30.3	28.6	24.7	21.2	18.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.7	13.2	13.7	13.2	11.5	10.9	10.8	11.6	13.5	14.0	12.1
		MCDBR	12.9	13.6	14.2	14.7	14.5	13.9	12.5	12.6	13.3	14.4	14.5	12.4
		MCWBR	8.7	8.3	7.0	6.6	5.7	4.6	4.1	4.1	4.8	6.5	8.3	8.6
	5% WB	MCDBR	10.6	10.9	11.3	11.0	11.9	11.3	10.8	10.6	10.7	10.2	10.7	10.2
		MCWBR	8.1	7.8	6.7	6.0	5.1	4.7	4.2	4.1	4.4	5.5	7.3	8.2
(40) Clear-Sky Solar Irradiance	taub		0.322	0.330	0.355	0.383	0.431	0.473	0.523	0.510	0.428	0.369	0.346	0.326
	taud		2.529	2.509	2.439	2.361	2.263	2.203	2.070	2.103	2.317	2.456	2.468	2.531
	Ebn at Noon		905	930	926	908	861	821	779	784	841	873	867	880
	Edh at Noon		85	95	110	124	137	145	165	158	123	99	90	81
(44) All-Sky Solar Radiation	RadAvg		2.88	3.46	4.64	5.74	6.30	6.12	5.97	5.53	4.95	4.25	3.25	2.49
	RadStd		0.19	0.43	0.40	0.45	0.44	0.48	0.38	0.34	0.40	0.51	0.29	0.27

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+100
(47)	Regional (45 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	+101

Nomenclature: See separate page